

ADF and KPSS have opposite null hypotheses

ADF H_0 = unit root $\rightarrow$ a small p means STATIONARY

KPSS H_0 = stationary $\rightarrow$ a small p means NON-STATIONARY

KPSS says stationary
 $p > 0.05$

KPSS says non-stationary
 $p < 0.05$

**ADF says
stationary**
 $p < 0.05$

STATIONARY
proceed — no differencing

DIFFERENCE-STATIONARY
difference once, then re-test

**ADF says
non-stationary**
 $p > 0.05$

TREND-STATIONARY
**detrend, or add a trend
regressor — do not difference**

NON-STATIONARY
difference and re-test

Difference as little as possible. Seasonal first, then regular. Over-differencing injects artificial negative lag-1 autocorrelation.